

Medical-LLM Supervision for Peer Retrieval in a Rare-Disease Community

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Abstract. People affected by rare diseases seek peers with shared experiences, but diagnoses are often incomplete and interaction histories too sparse for learning. On a non-profit peer platform with 3,584 profiles, we use a medical LLM, MedGemma 27B, as relevance judge: it grades 6.3 million profile pairs with an expert-informed rubric, and students learn these grades so that members, including newcomers, are matched from a single profile encoding. A three-part evaluation anchors teacher and students to a healthcare professional, to historical user ratings and to each other. The LLM agrees more closely with the professional than a pre-existing rule-based recommender (Spearman 0.78 versus 0.53), and its students agree as closely. For held-out newcomer users, a fine-tuned embedding retrieval model orders their rated peers better than the rules (NDCG@5 0.77 versus 0.69) and slightly better than the LLM itself; once a new profile is transcribed, it ranks a new member against the other 3,583 profiles in 8 ms instead of 1.7 h on one A100 GPU. It concentrates suggestions less than pairwise multilayer perceptron (MLP) scorers but more than the rules, so deployment should monitor exposure. A theory of change links these results to helpful peer contact and defines the outcome study that tests it.

Keywords: Peer recommendation · Rare diseases · Online health communities · LLM relevance judgments · Knowledge distillation

1 Introduction

Rare diseases collectively affect an estimated 263–446 million people worldwide [23], with individual communities widely dispersed [35]. Reaching a diagnosis can itself take years: a European survey of 6,507 respondents reported a mean of 4.7 years (median 0.8) from symptom onset to confirmed diagnosis [12]. Online communities offer practical knowledge and emotional support [9], but members must still find someone with whom an exchange would be useful.

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Matching on recorded diagnoses alone can miss comparable experiences, particularly when a diagnosis is missing or provisional. Symptoms and caregiving experiences may also matter.

We study a non-profit platform for patients, relatives and healthcare professionals (HCPs). Its current recommender, which we will call *deployed rules*, is a hand-written rule system that scores a pair from structured profile fields and disease-ontology relationships [4,13]. User ratings offer an alternative training signal, but cover few possible pairs and are absent for new members. Our approach instead asks a medical large language model (LLM) about potential conversational benefit. MedGemma 27B [30], an open medical LLM that is competitive with open models of similar size on medical benchmarks and runs locally, is the *teacher*; since scoring every candidate with it would take thousands of long generations per query, smaller *student* matchers, an embedding retrieval model and a pairwise multilayer perceptron (MLP), learn its grades and match new members from their profile alone (Fig. 1).

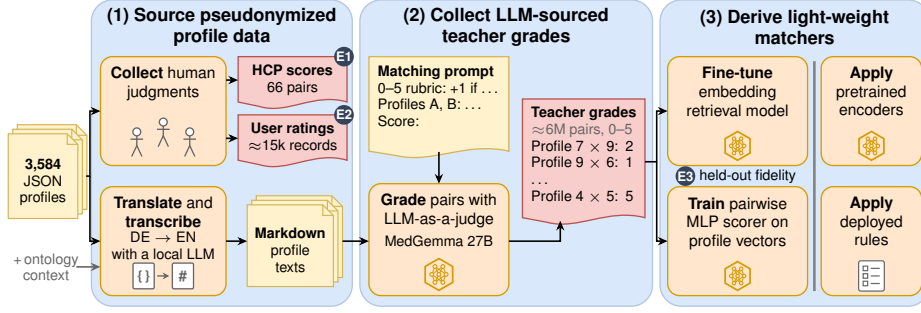


Fig. 1. Pipeline overview (MLP: multilayer perceptron). Teacher grades are the only student supervision; students encode the same Markdown texts. HCP scores (E1; pairs sampled across teacher grades) and user ratings (E2) evaluate teacher and students; E3 measures the students’ fidelity to the teacher on users excluded from training (Sect. 5). Example rows are placeholders.

We bring LLM relevance judging and distillation (Sect. 2) to peer discovery through medical and lived experience. We ask whether the teacher agrees with an expert (E1), aligns with user ratings (E2), and transfers its ordering to students (E3). Separating these questions responds to LLM-judgment concerns [8, 11]. Our contributions are (i) a three-part evaluation that anchors an LLM teacher and its students to an expert and to user ratings, with exact chance levels; (ii) evidence that a fine-tuned embedding retrieval model is as faithful to the teacher and as close to the expert as pairwise MLP scorers, beats the deployed rules on newcomers’ ratings, and spreads exposure more evenly; (iii) an exposure audit showing that students concentrate suggestions beyond what their labels imply; and (iv) a testable theory of change.

2 Related Work

LLMs as Relevance Judges. Prompted LLMs can label relevance as accurately as human labellers and have been deployed for large-scale labelling [34]; an open-source toolkit reproduces these results [37]. Others caution that such labels need human validation [8, 11] and that evaluating systems with labels from the model that trained them is circular [8, 32]; stratified sampling by the LLM’s label reduces the human effort for this validation [21]. We follow this advice: the teacher is anchored to an expert (E1) and to member ratings (E2), and agreement with the teacher (E3) is reported as fidelity, not as quality.

Synthetic Labels and Distillation for Retrieval. LLMs generate training queries for rerankers [6] and retrievers [7], their rankings can be distilled into smaller rerankers [33], and their graded relevance scores can train dense encoders directly [36]; knowledge distillation also transfers a slow ranker’s score margins to efficient students [16]. These methods rank documents for queries. We instead grade pairs of members of one community with an expert-informed rubric and distill the grades into one vector per profile with a ranking loss over graded labels [17], so that newcomers need neither interactions nor teacher calls.

Peer and People Recommendation. In a small cancer peer-mentoring study, participants rated sample posts as the most helpful profile element for deciding whether to contact a mentor [15]; a later field study examined recommendations through connection behavior [20]. Peer discovery is a social task, distinct from matching clinical cases for gene discovery, as in Matchmaker Exchange [25]. Earlier platform studies explored LLM profile augmentation [4] and feedback-trained matchers [2, 3]. Here, synthetic grades train students and ratings serve only for evaluation, so results are not directly comparable. People recommenders are often reciprocal: a match succeeds only if both sides accept it, so recommenders combine both users’ preferences [24, 26]. Rankings also concentrate attention on top-ranked people [5], which motivates constraining how exposure is allocated [31]; our exposure audit measures this concentration for learned peer-matching models.

3 Theory of Change: From Retrieval to Peer Support

Patients and relatives affected by rare diseases seek understanding and practical knowledge, including when diagnoses are missing or provisional. Profile-based discovery needs neither newcomers’ interaction histories nor behavioral labels for student training. Local LLM inference keeps profiles off remote services, and profile vectors avoid per-candidate LLM calls at serving time. Figure 2 places retrieval on the path to useful exchange.

From Ranked Profiles to Wanted Contact. Suitable peers must exist, be reachable and wish to engage. The Hit@10 of 0.90 of our fine-tuned (FT) embedding

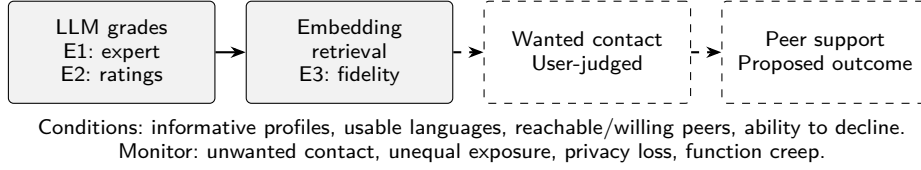


Fig. 2. Evidence-to-pathway map. Solid links are assessed here (E1–E3); dashed links are the targets of a consented, beneficiary-informed outcome study.

retrieval model, FT Arctic+cosine, means it retrieves a teacher-score- ≥ 4 peer for about 90% of evaluated test queries (Sect. 5.3), against 0.52 for random ordering and 0.72 for the deployed rules. Contact is also two-sided: reciprocal recommenders score each direction from that person’s own preferences [26], which a single symmetric similarity or expert score cannot. Contact therefore requires the ability to decline, informative profiles and usable languages.

Who Might Be Missed or Harmed? The rubric’s highest grades favor shared diagnoses (Sect. 4.1), potentially disadvantaging undiagnosed members. Demographic matching may reinforce homophily. The rubric rewards shared conditions at different stages, which can offer insight but also cause distress.

Better discovery can also increase unwanted exposure, and rankings may concentrate attention on a few members [5]: on the test lists, the pairwise MLP scorers give their most-exposed 10% of candidates 44% of top-ten slots, versus 26% under the deployed rules and 38% under FT Arctic+cosine (Sect. 5.4). Peers pushing lay diagnoses can be unwelcome, and professional or advocacy accounts may treat a group as serving their own agenda rather than its members’ [9]; the index could also be repurposed for marketing or recruitment. Responsible deployment requires purpose-limited access, control over visibility and contact, blocking/reporting, and separation from medical advice.

A future study should compare mutually accepted contact and helpfulness while monitoring unwanted contact, exposure and subgroup coverage. The evaluations below address the earlier step: whether profile judgments provide a useful retrieval signal.

4 From Profiles to Learned Matchers

4.1 Preparing Profiles and Grading Pairs

The dataset contains 3,584 profiles with diagnoses and symptoms coded as ontology identifiers, demographic fields and free text. The operator pseudonymized them before processing, and the LLM ran locally. Before transcription, each profile’s disease and symptom identifiers are resolved against the disease ontology, and these entries are added as context; MedGemma 27B then translates the

German JSON profiles into English and transcribes them into Markdown, which both teacher and students use.

The teacher assesses each unordered, non-self pair: $3,584(3,584 - 1)/2 = 6,420,736$ possible judgments. An expert-informed additive rubric awards up to five points: one if both users live with a rare disease and seek peer connection; a second for overlapping challenges, emotional goals or caregiving roles; a third for similar clinical characteristics such as shared symptoms, affected systems or disease trajectory; a fourth for experiential and medical complementarity, for example the same condition at different stages or medically related conditions; and a fifth only for the same or closely related diagnoses together with goals and life contexts that promise a mutually beneficial exchange. The prompt also asks the model to weigh demographics and disease ontologies.

The prompt requests examination, rubric application and justification within 800 words, then a score. Of 6,420,736 pairs, a random 2.0% were not graded and 2.4% fail schema filtering, leaving 6,136,498 valid grades (95.6%). They concentrate at 1–2 (84%), with 7.2% at 4–5, whereas the deployed rules concentrate near zero (Fig. 3(a)). Grading all pairs, not a shortlist, avoids preselecting candidates.

4.2 Learning an Embedding Retrieval Model and a Pairwise MLP Scorer

To avoid asking the teacher about every pair, the embedding retrieval model embeds a profile x_u as its profile vector $e_u = f_\theta(x_u)/\|f_\theta(x_u)\|_2$ and scores two profiles with $s(u, v) = e_u^\top e_v$. Each candidate’s profile vector can be reused across queries.

We use Snowflake Arctic Embed M v2.0 [38] as the encoder (**Arctic+cosine**). Inputs are truncated to 512 tokens, mean-pooled and ℓ_2 -normalized. Fine-tuning (**FT Arctic+cosine**) uses CoSENT [17] loss, which penalizes cosine orderings that contradict the graded labels rather than requiring a binary positive/negative threshold. Its scale is 20. Optimization uses AdamW with learning rate 10^{-5} and weight decay 10^{-2} , for 20 epochs of 264,000 pairs.

The second student, the pairwise MLP scorer, is a multilayer perceptron (MLP) on concatenated pairs of frozen profile vectors. It has a 512-unit hidden layer, ReLU and dropout of 0.1, and predicts the teacher’s grade using mean squared error. Training uses Adam, learning rate 10^{-4} , weight decay 10^{-5} , 200 epochs, and up to 10^6 pairs per epoch. We train it on pretrained and on fine-tuned profile vectors (**Arctic+MLP**, **FT Arctic+MLP**); a linear scorer without the hidden layer (**Arctic+linear**) is an ablation.

Grades are imbalanced, so each training epoch draws pairs with probability proportional to $1/(Kn_y)$, where n_y is the frequency of level y among the $K = 6$ levels; every level then contributes equally. Checkpoints are selected by full-list validation NDCG (epoch 4 of 20 for the encoder; epochs 54 and 166 of 200 for **Arctic+MLP** and **FT Arctic+MLP**). As validation and test share users, selection may favor the students slightly: the selected epochs exceed the median epoch’s validation NDCG@5 by about 0.01. Validation and test use all pairs, with

their natural grade distribution. Comparators are the deployed rules and two pretrained embedding retrieval models, Arctic+cosine and MSMARCO+cosine (MSMARCO BERT Base Dot v5,⁵ a Sentence-BERT model [28] trained on MS MARCO [1]).

4.3 Separating Training from Evaluation

Of the 3,584 users, 356 are held out. Validation and test share these users but not pairs: each split holds 30,153 pairs, and the test split covers 48% of the held-out users’ possible pairs, 8.4% of them teacher-high (teacher grade ≥ 4). Training uses exactly the 4,977,892 graded pairs among the other 3,228 users; all 1,098,300 cross pairs are excluded from training, and their rated pairs evaluate newcomers (Sect. 5.2; Fig. 3(b)).

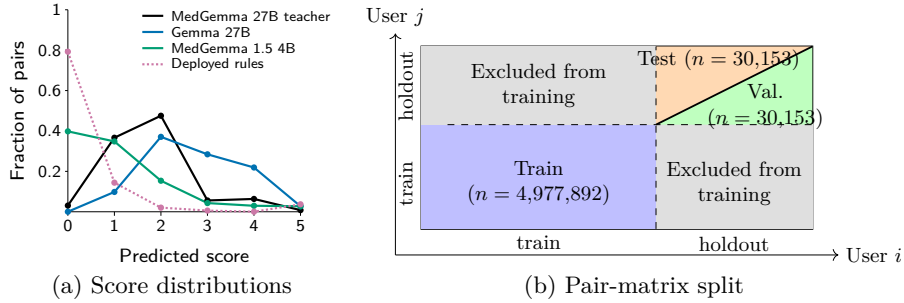


Fig. 3. (a) Score distributions of the deployed rules (all pairs they score) and of the LLM graders: the rules concentrate near zero, while the LLMs assign more intermediate and high grades. (b) Pairs of a training and a held-out user are excluded from training (their rated pairs evaluate newcomers); the holdout–holdout region is partitioned into validation and test pairs.

5 Three Views of the Evidence

5.1 E1: Does the Teacher Agree with an Expert?

One HCP saw parsed profiles, but not the teacher’s labels, and scored from 0 to 5 whether each pair might be interested in interacting. Like stratified validation of LLM relevance labels [21], we stratify by the teacher’s score: sampling eleven pairs at each level yields 66 judgments and covers the full score range. All 66 pairs and their 93 users come from the test split, so no student saw them in training. Because these are pairs rather than complete candidate lists, we use Spearman correlation.

⁵ <https://huggingface.co/sentence-transformers/msmarco-bert-base-dot-v5>

The teacher agrees more closely with this HCP than the deployed rules do: 0.78 versus 0.53 (Table 1), supporting its use as supervision. The paired difference of 0.25 has a 95% interval of 0.07–0.47 and stays between 0.18 and 0.28 when any one of the 93 users is left out. Distillation carries this agreement over to held-out users: students trained on teacher grades reach 0.80–0.84, whereas Arctic+cosine reaches 0.51 (fine-tuning it, FT Arctic+cosine, adds 0.31, interval 0.11–0.52). The students’ edge over the teacher (0.02–0.06) is within sampling error; their continuous scores also break the ties among the teacher’s six grades.

Table 1. Spearman agreement with the HCP on 66 teacher-stratified test pairs. Intervals: 10,000 bootstrap resamples of pairs within teacher-grade strata. FT: fine-tuned.

Scorer	ρ	95% CI
Deployed rules	0.53	0.33–0.70
MedGemma 27B teacher	0.78	0.67–0.87
Arctic+cosine (pretrained)	0.51	0.30–0.70
Arctic+MLP	0.84	0.75–0.91
FT Arctic+cosine	0.82	0.72–0.89
FT Arctic+MLP	0.80	0.69–0.88

The high grades illustrate a trade-off. Of eleven HCP-high pairs (score ≥ 4), nine are teacher-high and one is rules-high. The deployed rules miss ten at this threshold. Conversely, only nine of the 22 teacher-high pairs are HCP-high: the teacher grades more permissively than the HCP.

5.2 E2: Do the Grades Align with Users’ Historical Ratings?

Members also rate suggested peers: 15,356 feedback records cover 14,889 pairs, 13,697 of them with a valid teacher grade. For users with at least two such candidates, we rank them and compare their order with feedback using normalized discounted cumulative gain (NDCG), which emphasizes the top of the ranking.

On all rated pairs, MedGemma 27B exceeds the deployed rules at each cutoff (Table 2, upper block; NDCG@5 difference 0.06, 95% interval 0.05–0.07 over users). Rated lists are short (median six candidates for 1,915 users), so random order already reaches NDCG@5 0.56; both scorers clear it. We chose MedGemma 27B without consulting the ratings, so E2 is an independent check of the teacher; exploratory Gemma 3 27B [14] and MedGemma 1.5 4B [29] runs on a uniformly sampled 1% of pairs appear only in Fig. 3(a).

Students can only be checked on ratings of users they never saw. We therefore treat the held-out users as newcomers: each ranks the training users it rated or was rated by. These newcomer–training pairs were excluded from training (2,930 rated pairs from 201 newcomers; Table 2, lower block). FT Arctic+cosine exceeds the rules by 0.09 NDCG@5 (95% interval 0.06–0.12), and fine-tuning adds 0.07

Table 2. Ranking agreement with historical user feedback: NDCG over each user’s rated candidates; rows within a block rank the same lists. Newcomers: held-out users ranking the training users they rated or were rated by. Random order is the exact expectation. FT: fine-tuned. Differences in the text are computed before rounding.

Scorer	NDCG		
	@5	@10	@all
<i>All users (1,915)</i>			
Random order	0.56	0.62	0.69
Deployed rules	0.66	0.70	0.75
MedGemma 27B teacher	0.71	0.75	0.79
<i>Newcomers (201)</i>			
Random order	0.58	0.63	0.69
Deployed rules	0.69	0.72	0.76
MedGemma 27B teacher	0.75	0.77	0.80
MSMARCO+cosine	0.71	0.74	0.78
Arctic+cosine	0.71	0.75	0.78
Arctic+MLP	0.72	0.75	0.78
FT Arctic+cosine	0.77	0.80	0.82
FT Arctic+MLP	0.74	0.77	0.80

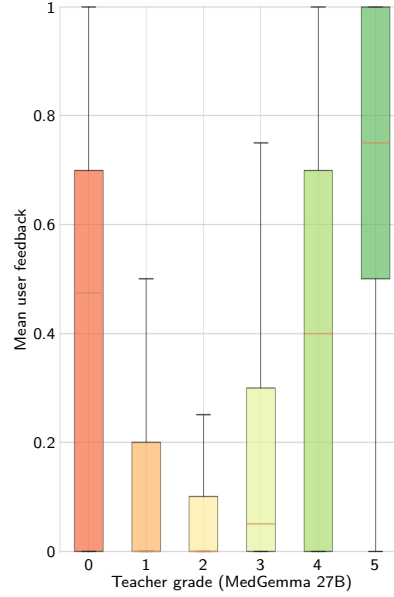


Fig. 4. Mean user feedback per pair by teacher grade. Grades 0–5 contain 134, 2,939, 5,921, 1,452, 2,494 and 757 pairs, respectively.

(0.04–0.09) over Arctic+cosine. It also edges the teacher (0.03, 0.01–0.05), possibly because the teacher’s six grades leave many ties; checkpoint selection may account for about 0.01 of this.

Figure 4 groups 13,697 pairs by teacher score, with feedback averaged within each pair. Higher score groups tend to have higher feedback, but the pattern is not monotonic: the 134 score-zero pairs have relatively high feedback. Inspection linked this exception to sparse profiles and possible offline acquaintance.

5.3 E3: Can a Smaller Model Learn the Teacher’s Ordering?

Does this signal survive distillation? On the teacher-labeled test split (356 users, a median of 169 candidates per query), NDCG measures ordering by teacher grade, while Precision (P) and Hit Rate (Hit) use grade ≥ 4 .

Table 3 shows that learning from these grades improves top-ranked agreement. Fine-tuning Arctic+cosine into FT Arctic+cosine raises NDCG@5 from 0.64 to 0.77, against 0.43 for random order, and improves NDCG@10 for 319 of the 356 test users (mean +0.13). Arctic+MLP already reaches 0.74; without its hidden layer, Arctic+linear falls to 0.57, below Arctic+cosine, because it ranks every query’s candidates alike (Sect. 5.4). Holding the scorer type fixed, the fine-tuned

Table 3. Teacher-label fidelity on the test split: how well each model reproduces the teacher’s ranking of candidates for users excluded from training, not how the users themselves would rank them. NDCG@5 is the primary top-ranked metric; P and Hit use teacher score ≥ 4 . Random order is the exact expectation. FT means fine-tuned.

Model	NDCG			P		Hit	
	@5	@10	@all	@5	@10	@5	@10
Random order (expected)	0.43	0.46	0.86	0.09	0.09	0.34	0.52
Pretrained embedder baselines (cosine)							
MSMARCO BERT Base Dot v5	0.60	0.62	0.90	0.30	0.26	0.67	0.74
Arctic Embed M v2.0	0.64	0.65	0.91	0.35	0.31	0.67	0.75
Existing task baseline							
Deployed rules	0.58	0.59	0.89	0.26	0.23	0.60	0.72
Our models							
Arctic+linear	0.57	0.60	0.90	0.22	0.20	0.50	0.64
Arctic+MLP	0.74	0.76	0.94	0.48	0.43	0.81	0.88
FT Arctic+cosine	0.77	0.78	0.94	0.54	0.48	0.84	0.90
FT Arctic+MLP	0.76	0.78	0.94	0.51	0.46	0.84	0.91

encoder’s NDCG@5 is about 0.13 higher under cosine and 0.02 higher under the MLP.

Full-list NDCG hardly separates scorers, since random order already reaches 0.86. The deployed rules beat random order (NDCG@5 0.58 versus 0.43) but trail the pretrained embedding retrieval models. FT Arctic+cosine matches FT Arctic+MLP (0.77 versus 0.76 NDCG@5) while reusing one vector per profile. E3 measures fidelity to the teacher, which by itself is no evidence of quality [8, 32]; E1 and E2 anchor the teacher to human judgments.

5.4 Who Gets Shown? Exposure on the Test Lists

Exposure asks who gets suggested, not whether a suggestion is good. We measure it on the test lists of Table 3: each of the 356 users excluded from training acts as a query over a median of 169 candidates (144–196), and every scorer fills a top ten, giving 3,560 suggestion slots. A user’s exposure is the number of slots they receive, averaging ties. It needs no labels and says nothing about individual match quality: it shows how evenly a scorer spreads its suggestions (Table 4, Fig. 5). The Gini coefficient is 0 if every user is suggested equally often and approaches 1 if a few users take all slots; Top 10% is the share of slots held by the most-suggested tenth of users (0.10 under equality); Never counts users in nobody’s top ten. Two diagnostics explain the differences: Overlap is the share of a scorer’s top ten that the deployed rules would also show, and the profile R^2 , defined next, measures how much a scorer’s ranking depends on who the candidate is, regardless of who is searching.

Following baseline estimates in recommendation [19] and the social relations model for dyads [18], we fit each scorer’s test scores as $s(u, v) \approx \mu + a_u + a_v$.

Table 4. Top-ten exposure on the test lists of Table 3 (356 users, median 169 candidates): how evenly each scorer spreads its suggestions across users. The exposure columns use no labels and do not measure match quality; NDCG@5 repeats Table 3 and measures agreement with the LLM teacher’s grades, not the HCP’s. Columns are defined in Sect. 5.4; ties are averaged and random shows expected exposure. Bold marks the best scorer (highest NDCG, least concentration). FT means fine-tuned.

Model	NDCG		Top-ten exposure		Diagnostics	
	@5	Gini	Top 10%	Never	Overlap	R^2
Random order (expected)	–	0.03	0.11	0	0.06	–
Teacher-grade order	–	0.41	0.27	0	0.15	0.30
Pretrained embedder baselines (cosine)						
MSMARCO BERT Base Dot v5	0.60	0.48	0.33	14	0.14	0.31
Arctic Embed M v2.0	0.64	0.42	0.27	12	0.16	0.28
Existing task baseline						
Deployed rules	0.58	0.43	0.26	8	–	0.26
Our models						
Arctic+linear	0.57	0.91	0.91	282	0.10	0.95
Arctic+MLP	0.74	0.65	0.44	82	0.17	0.60
FT Arctic+cosine	0.77	0.53	0.38	21	0.19	0.63
FT Arctic+MLP	0.76	0.63	0.44	67	0.17	0.56

Each member has one effect a_u , how highly the scorer rates u with anyone, and the effects sum to zero. We minimize the squared error over all observed test pairs by alternating updates, setting each a_u to the mean residual of u ’s pairs until the effects converge. The profile $R^2 = 1 - \sum (s - \hat{s})^2 / \sum (s - \bar{s})^2$ is the share of score variance that this fit explains. Near 0, scores depend on who is paired with whom, so every member receives a different ranking; near 1, a candidate scores almost the same for every query, so all members see the same people first. Order-dependent MLP scores are approximated by effects that are the same in both roles.

Fidelity and concentration do not rise together (Fig. 5(b)). Ordering by the teacher’s own grades spreads exposure as evenly as the deployed rules (Gini 0.41 versus 0.43), so the students’ concentration does not come from their labels. FT Arctic+cosine is the most faithful student and concentrates exposure less than the pairwise MLP scorers (Gini 0.53 versus 0.63–0.65): the MLP scorers never show 67–82 of the 356 candidates, FT Arctic+cosine 21 and the rules 8.

Arctic+linear shows the mechanism in its extreme form. Its score is a weighted sum of the two profile vectors, so for a given query only the candidate’s part of the sum varies: every query ranks its candidates in nearly the same order, and the same few candidates fill most top-ten slots (Gini 0.91; 282 of 356 candidates never shown). More generally, concentration rises with the share of score variance that per-profile effects explain (R^2 from 0.26 for the rules to 0.95; Spearman 0.86 across the seven scorers of Table 4). The teacher’s grades have an R^2 of only

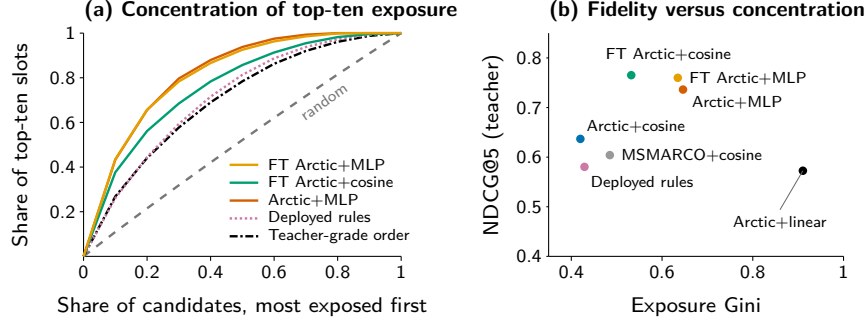


Fig. 5. Exposure on the test lists (356 users). (a) Share of top-ten slots held by the most-exposed candidates (deciles; ties averaged, random expected); pretrained Arctic+cosine lies on the rules’ curve. (b) Teacher NDCG@5 (Table 3) against exposure Gini for each scorer.

0.30, so the students amplify per-profile effects beyond their labels (0.56–0.63). Candidates that appear in the top ranks of many queries are known as hubs in high-dimensional data [27]; here they emerge through training on the teacher’s grades. Every scorer shares at most 19% of its top ten with the deployed rules, so replacing them would change most suggestions that members see.

6 Implications for Deployment and Evaluation

Serving Workload. Table 5 breaks serving into stages. Every path starts with one LLM transcription per profile. The teacher then needs one LLM generation per pair; the students need one encoding per profile and one vector operation per pair, about $6 \cdot 10^{10}$ times fewer operations per pair. A new member is thus ranked against the other 3,583 in 8 ms after a 6.4 s transcription, instead of 1.7 h, and grading all 6.4 million pairs takes about 125 GPU-days, against 1.6 h of transcription and 10 s of encoding. FT Arctic+cosine matches the fidelity of FT Arctic+MLP (Table 3), so serving needs a 1.2 GB encoder and 11 MB of profile vectors; only transcription needs the 27B model, once per profile. Teacher labels, needed only for training, are the main cost.

Scope and Next Steps. One HCP judged 66 teacher-stratified test pairs; a second blinded annotator would test the reliability of these judgments. Feedback covers only candidates the deployed rules had shown, so E2 compares scorers within those lists; newcomers’ lists are short (median five rated candidates). The rubric is not yet beneficiary-validated. The exposure audit covers the 356 test users rather than the whole platform. Diagnosis and completeness breakdowns, repeated runs and a beneficiary study of the contact-to-support pathway complete the agenda.

Table 5. Serving cost per stage on one NVIDIA A100 (80 GB). Teacher and students start from one transcription per profile. Times are measured: single requests for transcribing and encoding a new member, batched runs otherwise. FLOPs are estimated as $2 \times \text{parameters} \times \text{tokens}$ for the LLM and the encoder (mean tokens per request; the encoder reads one Markdown profile) and counted exactly for the scoring heads; memory is that of the weights or of the profile vectors, which have no parameters (for the MLP, 3 MB of weights and 15 MB of first-layer products precomputed per profile in place of its vectors, so that a pair needs only a 512-dimensional addition and dot product). LLM FLOPs count input and output tokens alike, but output tokens are generated one at a time, so measured times are the better guide to cost.

Stage	Model (memory)	Params	Unit	Tokens in/out	FLOPs/unit	New member		All pairs	
						units	time	units	time
Transcription	MedGemma 27B (54 GB)	27 B	profile	3,912/212	$2.2 \cdot 10^{14}$	1	6.4 s	3,584	1.6 h
<i>Teacher</i>									
Grading	MedGemma 27B (54 GB)	27 B	pair	959/824	$9.6 \cdot 10^{13}$	3,583	1.7 h	6.4 M	125 d
<i>Students</i>									
Encoding	Encoder (1.2 GB)	305 M	profile	212/–	$4.8 \cdot 10^{10}$	1	8 ms	3,584	10 s
Cosine scoring	Profile vectors (11 MB)	–	pair	–	$1.5 \cdot 10^3$	3,583	0.09 ms	6.4 M	0.5 ms
MLP scoring	MLP (18 MB)	0.79 M	pair	–	$2.0 \cdot 10^3$	3,583	0.15 ms	6.4 M	0.09 s

Ethics and Data Governance. Scores are peer-discovery suggestions, not medical advice. The algorithms were developed for the platform’s matching service. The platform was developed with a university hospital’s data protection officer and in consultation with the local ethics committee; under the committee’s statutes no formal vote was required, as the work is neither a clinical trial of a medicinal product nor a clinical investigation of a medical device [13]. The operator pseudonymized all profiles before processing; we ran all LLM inference locally on academic infrastructure in Germany. Because even pseudonymized health attributes and embeddings can enable re-identification or text recovery [10, 22], profiles, embeddings and labels will not be released.

7 Conclusion

Where interaction data are too sparse, a medical LLM can supply the relevance judgments that peer retrieval needs. In this rare-disease community, MedGemma 27B agrees more closely than the deployed rules with an expert and with historical ratings. Its grades teach the embedding retrieval model FT Arctic+cosine, which raises fidelity from 0.64 to 0.77 NDCG@5 for users excluded from training. This model ranks these users’ rated peers better than the deployed rules (0.77 versus 0.69 NDCG@5) and agrees with the expert as closely as the teacher. The pipeline is also efficient end to end: on one A100 GPU, a new member is transcribed and ranked against the other 3,583 in 6.4 s instead of 1.7 h with the teacher, and all 3,584 profiles in 1.6 h instead of 125 GPU-days. FT Arctic+cosine concentrates exposure less than the pairwise MLP scorers, but every learned model would replace most current suggestions, which deployment should

monitor. Next comes a consented study of mutually wanted, helpful contact and its harms.

Use of Generative AI. Generative models are part of the method: MedGemma 27B transcribes and translates profiles and produces the teacher grades (Sect. 4.1). Separately, OpenAI Codex and Anthropic Claude Code assisted with drafting text, editing, figure and analysis code, and review. The authors reviewed all generated text, code and reported numbers.

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